

FPGA emulation card for the nanoSoC Ethernet chiplet slot

Draft for review | 2026-09-30 | revision 2: K26I slot card

For the bring-up and PCB engineers. Device: nanoSoC Ethernet chiplet rc4-20260829 (TSMC 65 nm). Interface: slot contract v1, 120-position HSEC8.

Verdict: build a Kria K26I slot card (CARD_ID 001). It plugs into the same slot as the bonded-die card, so the two are interchangeable, as the user requires.

- **Why the K26I:**

- It is the same XCK26 device as the proven KR260 build. The whole chiplet (58k LUTs, including TideLink) uses 51 %.
- Its 69 HDIO at 3.3 V carry all 48 slot signals. Banks 43 + 44 hold exactly 48.
- The nanoSoC FPGA Toolkit already supports zynqmp.
- It is in stock: 112 at Mouser UK, £439.21, on 2026-09-30. The K26C had 0.

The K26I differs from the K26C only in temperature grade and shock/vibration qualification (DS987 v1.2 tables 1, 22, 25). Build for xck26-sfvc784-2LV-i.

- **Now:** develop the emulator bitstream on the lab's KR260s, which use the same device.

- Undo the QSPI/HOSTIO4 tie-offs.
- Take CLK from a pin.
- Use the silicon boot ROMs.
- Start from the HAPS-SX board top.

- **Optional:** a KR260 + paddle adapter (36 IO; no TideLink). Build it only if the motherboard is ready well before the K26 card.

- **Effort^{est}:**

- card: 20–26 engineer-days plus 3–4 weeks of fabrication and assembly;
- bitstream: 16–22 engineer-days, in parallel on a KR260;
- **total: about 36–48 engineer-days, 2–3 months elapsed.**

Table 1. Options. “Fits” = post-route LUT use of 75 % or less for the whole chiplet (58k LUTs). ^{est} marks my estimates.

Option	Fits?	3.3 V IO	Same slot as die card?	Stock, price (2026-09-30)	Decision
K26I SOM on slot carrier	51 %	69 HDIO	yes	112 at Mouser UK, £439	Chosen. Proven device; all 48 signals; toolkit supports it.
K26C SOM on slot carrier	51 %	69 HDIO	yes	0 in stock, £353	Same card. Use it if stock returns; 0–85 °C only.
Trenz TE0712 A7-200T on carrier	43 %	158 HR	yes	0 in stock, €289 net	Fallback. Lower power. The toolkit's bin_style is {zynq7, zynqmp, none} (fpga/fpga-toolkit/part/pack_schema.tcl:494), so booting an Artix-7 from SPI flash needs an extension.
K24 SOM on carrier	82 % of 70.6k	23 HDIO	yes	—	Rejected. Too few 3.3 V IO; the 1.8 V HPIO would need level shifters (DS985).
Kria on the motherboard	51 %	69	no	—	Rejected. Breaks interchangeability.
KR260 + paddle adapter	51 %	36 used	yes (cabled)	owned	Optional. Only if the motherboard is ready well before the card. No TideLink or I2C.
Nexys Video + FMC cable	43 %	68 LA	cabled	\$711	Not pursued. A bench cable, not a slot card.

1 Design size and the existing FPGA builds

Table 2. Measured FPGA resource use (BRAM in 36 Kb tiles). Sources are under the table.

Build	Device	LUT	FF	BRAM	DSP	SoC clock, slack
[1] KR260 “landed”, whole design	xck26-2LV	59,851	54,337	32.5	7	25.0 MHz, +12.37 ns
of which nanosoc_eth_chiplet		58,031	52,414	32.5	7	
SoC (u_soc)		28,212	19,342	28.5	6	
TideLink		23,672	27,673	4	1	
TideChart + APB bridges + D2D decode		6,170	5,364	0	0	
[2] HAPS-SX, PHY=rmi	xcvu19p-2	57,617	51,503	41	7	25 MHz; hold fails
[3] PYNQ-Z2 multicore SoC, no TideLink	xc7z020-1	27,958	19,827	31.5	6	25 MHz, +0.58 ns

[1] fpga/build/landed/reports/utilization_impl.rpt:36, 41, 109, 125; utilization_hier_impl.rpt:351, 357, 1041, 1033, 1032, 1509, 356; timing_summary.rpt:197, 215. [2] ~/SoCLabs/HAPS-work/fpga/haps-sx/build/vivado_out/post_route_util.rpt:40, 113; README.md:19-27. [3] nanosoc-multicore-system/imp/fpga/project/pynq-z2/nanosoc_multicore_project.runs/impl_1/nanosoc_multicore_design_wrapper_utilization_placed.rpt:35, 40, 106; ..._timing_summary_routed.rpt:154, 169.

- **Size.** The whole chiplet is about 58k LUTs, and TideLink + TideChart are about 30k of that. The “131k” figure for the KR260 is its primitive-cell count. On the K26 (117,120 LUTs) it is 51 % full; on the K24 (70,560 LUTs, DS985) it would be 82 %. Memory is small: 32.5 of 144 BRAM tiles.
 - **Clock.** Every FPGA build runs the SoC at 25 MHz. On the K26 there is +12.4 ns of slack at 25 MHz, so about 35 MHz may be possible^{est}. The RMII domain runs directly from the 50 MHz REF_CLK with 18.5 ns of slack (timing_summary.rpt:211).
 - **The KR260 build cannot go into the slot as it is.**
 - The SoC clock and reset come from the PS: PLO 100 MHz through a clk_wiz to 25 MHz (tidelink/fpga/targets/kr260-eth-chiplet/tidelink_design.tcl:95, 108, 166).
 - QSPI, SPI and HOSTIO4 are tied off in the packaged IP wrapper (tidelink/fpga/vivado_ip/nanosoc_eth_chiplet_vivado_wrapper.v:257-269).
 - The eth ROM is smoke_remap, not silicon’s stage0. 172 of 320 words differ, and stage0 copies from QSPI flash and halts without it (docs/bringup/ASIC_FIRST_SILICON_HOSTIO_PROCEDURE.md:123).
 - IMEM is preloaded with RAM_PRELOAD (tidelink/fpga/vivado_ip/nanosoc_eth_chiplet_filelist.tcl:171-225).
 - **The HAPS-SX board top already has what the slot needs.**
 - It wires QSPI, HOSTIO4, SWD and RMII (HAPS-work/fpga/haps-sx/vsrc/nanosoc_eth_chiplet_haps_sx.sv:95-156).
 - It loads the rc4 silicon eth ROM; CPU1’s ROM already matches silicon (HAPS-work/fpga/haps-sx/Makefile:206-215).
 - It times QSPI SCK as HCLK/2, like silicon at boot (README.md:305-310).
- Its hold failures (README.md:19-27) must be re-checked on the K26.

2 K26I slot card: what to build

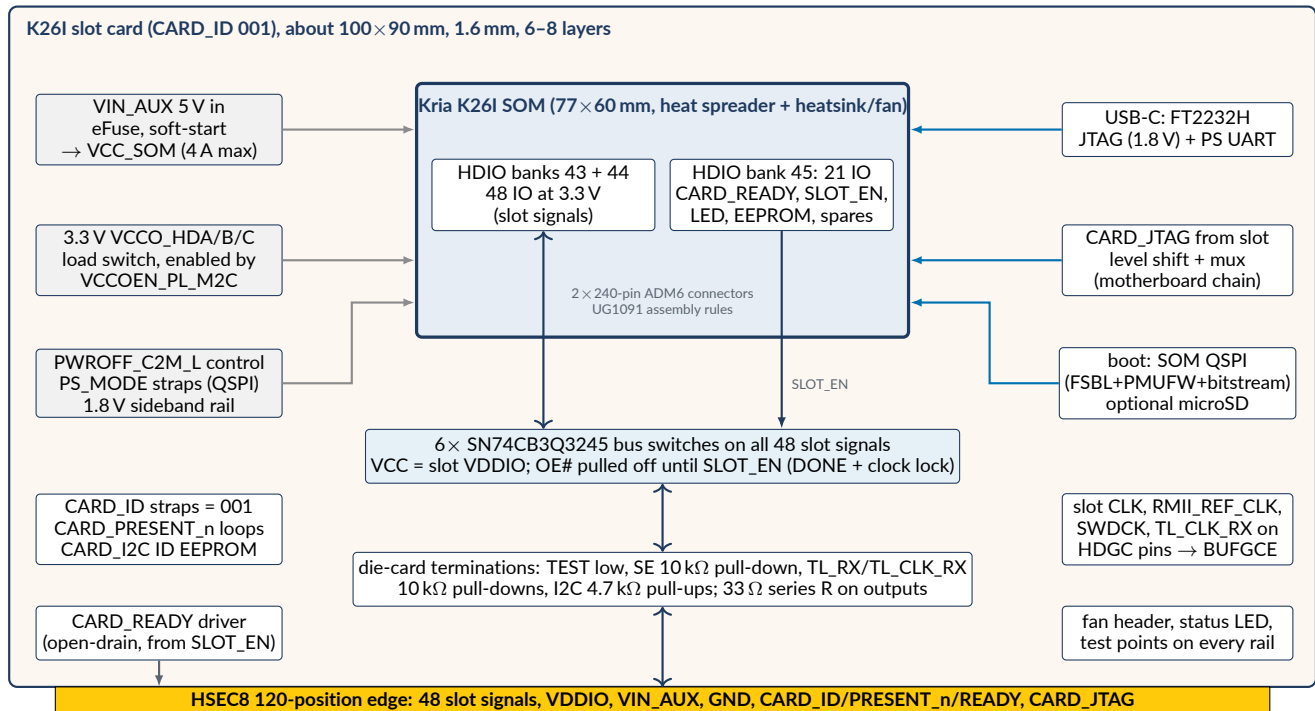


Figure 1. Block diagram of the K26I slot card. Slot signals run from HDIO banks 43 and 44, through loff-rated bus switches powered from slot VDDIO, to the edge fingers. The die-card terminations sit on the slot side of the switches, so the slot nets look the same whichever card is fitted.

Table 3. Carrier requirements. “DS987” and “UG1091” items were read from AMD documents; the rest are design choices for this card.

Area	Requirement	Source
SOM power	VCC_SOM is 5 V (4.75–5.25 V, 50 mV p-p max noise), 4 A max. Fed from slot VIN_AUX through an eFuse with soft-start. PWROFF_C2M_L (active low, pulled up to 5 V on the SOM) powers the SOM down and, when released, starts its power-on sequence; the carrier drives it from a power-on-reset / host-controlled signal.	DS987 table 17 and “Power Management Signals”; UG1091 “Power Sequencing”
HDIO bank power	The carrier supplies VCCO_HDA/HDB/HDC (1.2–3.3 V, 1 A each) and must enable them only after the SOM asserts VCCOEN_PL_M2C. Here: a 3.3 V rail from VIN_AUX through a load switch driven by VCCOEN_PL_M2C. The VCCO tolerance at the connector is +3%/–2%. VCCOEN_PS_M2C gates the PS-peripheral rails (UART, microSD) the same way.	DS987 table 17; UG1091 “PL Power Up Behavior”
Connectors	2 × Samtec ADM6-60-01.5-L-4-2-A (carrier side; 0.635 mm, 4-row, 240 positions each). UG1091 gives placement, distance-tolerance, stencil and reflow rules: use an assembler who has built Kria carriers.	UG1091 “SOM Connector Overview”
Boot	The SOM holds 512 Mb QSPI and 16 GB eMMC. PS_MODE[3:0] are pulled up on the SOM; strap 0010 (QSPI) with 0 Ω to GND. Own BOOT.BIN (FSBL + PMUFW + bitstream) in QSPI, so no Linux is needed. Optional microSD on MIO.	DS987 table 6, “Sideband Signals”
Debug	SOM JTAG pins are 1.8 V sideband. FT2232H USB (JTAG + PS UART) on a USB-C port, level-shifted. The slot CARD_JTAG (VREF = VDDIO) is buffered and muxed into the same chain.	DS987 “Sideband Signals”; contract v1
Slot IO isolation	6 × SN74CB3Q3245 (8-bit, 4 Ω, 500 MHz, loff) on all 48 signals. VCC comes from slot VDDIO. OE# is pulled to “off” until the bitstream drives SLOT_EN (DONE + clock lock), which also releases CARD_READY. Nothing reaches the slot during configuration or while VDDIO is off.	T1 product page; contract v1
Terminations	The die-card set on the slot side: TEST low, SE 10 kΩ pull-down, TL_RX[7:0]/TL_CLK_RX 10 kΩ pull-downs, I2C 4.7 kΩ pull-ups, 33 Ω series R on outputs.	contract v1
Identity	CARD_ID straps 001, both CARD_PRESENT_n loops, CARD_READY open-drain, CARD_I2C ID EEPROM, status LED.	contract v1
Pins and clocks	48 slot signals on HDIO banks 43 + 44 (24 + 24). Card control on bank 45 (21 IO). CLK, RMII_REF_CLK, SWDCK and TL_CLK_RX on HDGC pins.	DS987 pin table
Thermal, mechanics	The SOM (77×60×10.9 mm, 58 g) has a heat spreader, and the cooling solution attaches to it. Fit a heatsink + fan (fan header). Card up to about 100×90 mm; the motherboard provides a keep-out volume and a card guide or bracket; same keying as the die card; no hot-plug.	DS987 tables 22–23; contract v1

Table 4. Card effort (my estimates, one experienced PCB engineer).

Item	Effort ^{est}
Schematic: power and sequencing, SOM connectors, bus switches, JTAG/UART, boot straps, terminations, ID	6–8 d
Layout: 6–8 layers, ADM6 fan-out, 1.6 mm paddle with hard-gold HSEC8 fingers, heatsink keep-outs	8–10 d
Fabrication and assembly (fine-pitch connectors, gold fingers)	3–4 weeks elapsed
Carrier bring-up: rails, PWROFF / VCCOEN sequence, JTAG, QSPI boot, CARD_READY	4–6 d
Mechanical: heatsink/fan choice, card guide agreed with the motherboard	2 d
Card total	20–26 d + 3–4 wk

3 RTL and bitstream work (start now on a KR260)

Item	Content	Effort ^{est}
Board top	Fork the HAPS-SX top for the XCK26. Keep the PS only for configuration and debug. Remove the PS AXI-to-AHB backdoor, which is tied idle on silicon, and the tie-offs in <code>nanosoc_eth_chiplet_vivado_wrapper.v:257-269</code> .	3–4 d
Clocking	Slot CLK on an HDGC pin. At 25 MHz (contract v1 option) drive a BUFGCE directly with no MMCM, so CLK gating and NRST release match silicon. At 50/100 MHz use an MMCM $\div 2 / \div 4$ and hold reset until lock. The HDGC→MMCM route must be checked in Vivado, because the MMCMs sit in the HP columns ([unconfirmed]). RMII_REF_CLK drives the MAC domain directly.	2–3 d
Silicon ROMs	eth ROM = <code>eth_ss_bootrom.rc4.bintxt</code> (as <code>HAPS-work/fpga/haps-sx/Makefile:206-215</code>); CPU1 ROM already matches silicon. Drop RAM_PRELOAD. Firmware built for 25 MHz (<code>NS_INCR = 40</code> , <code>NANOSOC_SYS_CLK_FREQ_HZ</code>).	1–2 d
Pad wrapper	<code>nanosoc_eth_chiplet_pads_fpga.v: mirror</code> ASIC/tech_wrappers/tsmc65/nanosoc_eth_chiplet_pads.v with IOBUF and PULLUP/PULLDOWN matching rc4's pull states. Keep <code>nanosoc_eth_chiplet_bscan (:372-384)</code> so IDCODE, BYPASS and EXTEST work from the motherboard JTAG header.	3–5 d
Board packs	<code>fpga/board/kr260-slotdev/board.tcl</code> (KR260 pins, for development now) and <code>fpga/board/k26i-slot/board.tcl: part xck26-sfvc784-2LV-i, bin_style zynqmp, deploy_style jtag</code> , SOM-only board_part (the K26I board-file name is [unconfirmed]). Pin XDC for 48 + control signals.	2–3 d
Bring-up against the motherboard	Flash-boot from the motherboard QSPI with a <code>mkflash.sh</code> image; HOSTIO4 suite; RMII ping; Tidelink register reads with the link down; CARD_READY / NRST sequencing.	5 d

4 Slot contract v1: changes adopted

These supersede v0. All are now part of the contract, not proposals.

- Card types:** 000 die card, **001 K26I FPGA card**, 010 packaged die, 111 empty. Every card uses the same 120-position HSEC8 and the same keying. No hot-plug: all rails off before inserting or removing a card.
- VIN_AUX:** 5 V at 4 A or more continuous, on at least 6 contacts (HSEC8 is about 2.8 A per pin **[snippet]**, derated). Supplied through an eFuse with an INA228. The die card leaves it unconnected.
- VDDIO** is on for every card type. The FPGA card uses it to power its slot-side bus switches.
- CARD_READY:** open-drain, card to motherboard, pulled up on the motherboard. The FPGA card releases it after configuration and slot-IO enable; the die card straps it ready. The host MCU waits for it before releasing NRST.
- Card JTAG on spare pins:** TCK, TMS, TDI, TDO and VREF. The FPGA card also keeps its own USB-JTAG/UART.
- 25 MHz CLK option** on the motherboard oscillator, alongside 50/100 MHz with OE.
- VDD_CORE dummy load (DNP)** on the *motherboard*, so the core supply can be tested with the FPGA card fitted.
- Rules for every card:** die-card terminations and series resistors; the FPGA card isolates its slot IO with bus switches until configured.
- Mechanics:** FPGA card up to about 100×90 mm; a keep-out volume and card support on the motherboard.

5 Open questions

- The HDGC→MMCM route on the K26 at 50/100 MHz. If it fails, run the FPGA card only with the 25 MHz CLK.
- K26 power draw with this design. No `report_power` exists; DS987 gives only the 4 A maximum. Measure it on a KR260 to size the heatsink and confirm the VIN_AUX margin.
- Board files for a custom K26I carrier: the SOM-only board_part name, and whether the factory QSPI boot firmware expects a carrier ID EEPROM. The plan is to write our own BOOT.BIN.
- The KR260 bank 45 (PMOD1/2) fault (`ASIC_FIRST_SILICON_HOSTIO_PROCEDURE.md:165`): is it on the SOM or on the carrier? The card puts only control signals on bank 45.
- The HSEC8 per-pin current rating (**[snippet]**) and the Mouser K26I stock figure (from contract v1) should be re-read from the datasheet and distributor page before ordering.

Sources

Repository (read-only; paths cited inline; HAPS-work = ~/SoCLabs/HAPS-work). Slot contracts v0 and v1: bring-up research notes, 2026-09-30.

Web (fetched 2026-09-30 by me or by two research agents; **[snippet]** means seen in a search result only):

- AMD DS987, Kria K26 SOM data sheet v1.2 (tables 1, 6, 17, 22, 23, 25; sideband and power-management signals): <https://docs.amd.com/v/u/en-US/ds987-k26-som> (copy read: <https://www.sundance.com/wp-content/uploads/docs/k26-som-Data-Sheet.pdf>)
- AMD DS985, Kria K24 SOM (70,560 LUTs; 23 HDIO, 56 HPIO): <https://docs.amd.com/r/en-US/ds985-k24-som/Programmable-Logic>;
- AMD UG1091, Kria carrier card design (connectors, power sequencing, PL power-up): <https://docs.amd.com/r/en-US/ug1091-carrier-card-design/SOM-Connector-Overview>;
- AMD UG1092, KR260 interfaces: <https://docs.amd.com/r/en-US/ug1092-kr260-starter-kit/Interfaces>
- K26C DigiKey (0 stock, 26-week lead): <https://www.digikey.com/en/products/detail/amd/SM-K26-XCL2GC/13985266>
- Trenz TE0712 TRM and shop: <https://wiki.trenz-electronic.de/display/PD/TE0712+TRM>;
- <https://www.trenz-electronic.de/en/Products/Trenz-Electronic/TE07XX-Artix-7/TE0712-Artix-7/>
- Nexys Video (FMC LPC, VADJ): <https://boards.fpgadeveloper.com/boards/Nexys-Video>; DigiKey 410-316: <https://www.digikey.com/en/products/detail/digilent-inc/410-316/5456481>
- AMD DS180, 7 Series overview: https://docs.amd.com/v/u/en-US/ds180_7Series_Overview
- TI SN74CB3Q3245: <https://www.ti.com/product/SN74CB3Q3245>
- Samtec HSEC8-DV current rating **[snippet]**: https://www.mouser.com/datasheet/3/85/1/hsec8_dv.pdf
- Digilent Pmod interface specification, and TI AN-991 on ribbon cable (for the optional adapter): https://cdn.hackaday.io/files/21333912711072/Digilent-Pmod_%20Interface_Specification.pdf; <https://www.ti.com/lit/an/snla043/snla043.pdf>

K26l price and stock (Mouser UK, £439.21, 112 units) are taken from slot contract v1 and were not re-fetched. Effort figures are estimates, not measurements.